

# William Kwako

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wkwako.github.io

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## EDUCATION

M.S., Computer Science, Georgia Tech, Atlanta, GA

Expected graduation date: August 2026

B.A., Physics, Willamette University, Salem, OR

May 2019

## PROFESSIONAL EXPERIENCE

**Technical Solutions Engineer** – Epic Systems Corporation

December 2019 – December 2022

- Supported clients with deployment and maintenance of healthcare software in the Claims and Remittance department
- Headed the Debugging Task Force; responsibilities included teaching new hires how to use internal debugging tools, revising the curriculum, and delegating assignments to task force members
- Conducted quality assurance on multiple development projects using PQA framework
- Created searches with M to retrieve information from Healthcare IT databases for issue resolution and data analysis

**Teacher's Assistant** – Willamette University, Physics Department

January 2019 – May 2019

- Guided Students through computational and experimental Introductory Physics 2 laboratory exercises
- Answered questions about class material and evaluated students' performance

**Intern** – Willamette University, Physics Department

May 2018 – July 2018

- Analyzed frequency and location of baryon acoustic oscillations in 2MRS
- Located two BAOs with signals and characteristics agreeing with results of previous studies
- Worked closely with professor and peers to build computer for parallel GPU processing

**Volunteer** – UCSB Department of Cosmology

July 2017

- Assisted research staff in measuring the maximum intensity of light receivable by photodiodes
- End goal of sending small satellites to Proxima Centauri propelled by lasers and a solar sail system

## PROJECTS

- Improving Students' Code Comprehension Ability Through LLM-Generated Practice Problems  
*Educational technology project*
  - Built a webapp that leverages LLMs to generate programming exercises, aiming to improve students' code comprehension skills
  - **Tools:** Python, Django, JavaScript, HTML/CSS, OpenAI and Anthropic APIs
- Ravens' Progressive Matrices Solver  
*Knowledge-based AI project*
  - Created a Python program that applies computer vision and pattern analysis to automatically solve visual reasoning puzzles
  - **Tools:** Python
- Looking for the BAO Signal in the 2MRS Using the Wavelet Transformation  
*Senior Thesis, Willamette University*

- Applied wavelet transformations to identify baryon acoustic oscillations (BAOs) in galaxy distribution data from the 2MASS Redshift Survey (2MRS)
- **Tools:** Python
- Squire of the Crescent Realm  
*Video game development project*
  - Collaborated in a development team to design and implement a 3D adventure game using Unity and C#
  - **Tools:** C#, Unity, Git

## POSTERS AND PRESENTATIONS

- *Improving Students' Code Comprehension Ability Through LLM-Generated Practice Problems*  
Poster Presentation – National Council on Measurement in Education (April 2026)
- *Looking for the BAO Signal in the 2MRS Using the Wavelet Transformation*  
Oral Presentation: Willamette-Linfield intercollegiate symposium (May 2019)
- *Looking for the BAO Signal in the 2MRS Using the Wavelet Transformation*  
Poster Presentation: 223<sup>rd</sup> Meeting of the American Astronomical Society (January 2019)
- *Looking for the BAO Signal in the 2MRS Using the Wavelet Transformation*  
Poster Presentation: Murdock College Science Research Conference (November 2018)

## TECHNICAL SKILLS

### Programming

- Python
- Java
- M
- SQL
- C#
- CSS
- HTML

### Software

- Microsoft 365 Suite
- LaTeX
- Git
- Unity
- Android Studio

### Hardware

- Building computers
- Arduino-based electronics

## HONORS AND AWARDS

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|-------------------------------------------------------------------------------------|--------------------------|
| Jason Lee Scholarship                                                               | August 2015 – May 2019   |
| Willamette University College Honors                                                | February 2016 – May 2019 |
| Murdock Charitable Trust College Research Program for the Natural Sciences (\$5500) | June 2018 – August 2018  |